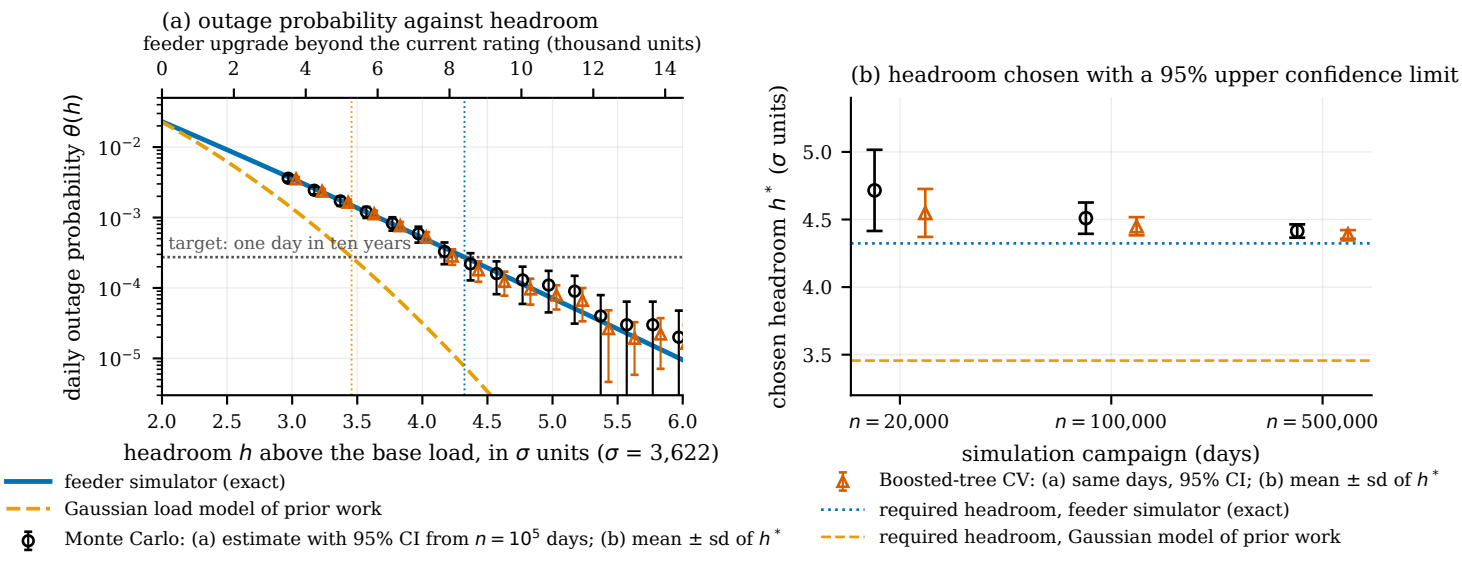
(a) outage probability against headroom
feeder upgrade beyond the current rating (thousand units)
0
2
4
6
8
10
12
14
daily outage probability $\theta(h)$
10^{-2}
10^{-3}
10^{-4}
10^{-5}
target: one day in ten years
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
headroom h above the base load, in σ units (σ = 3,622)
(b) headroom chosen with a 95% upper confidence limit
chosen headroom h^* (σ units)
5.0
4.5
4.0
3.5
$n = 20{,}000$
$n = 100{,}000$
$n = 500{,}000$
simulation campaign (days)
feeder simulator (exact)
Gaussian load model of prior work
Monte Carlo: (a) estimate with 95% CI from $n = 10^5$ days; (b) mean ± sd of h^*
Boosted-tree CV: (a) same days, 95% CI; (b) mean ± sd of h^*
required headroom, feeder simulator (exact)
required headroom, Gaussian model of prior work